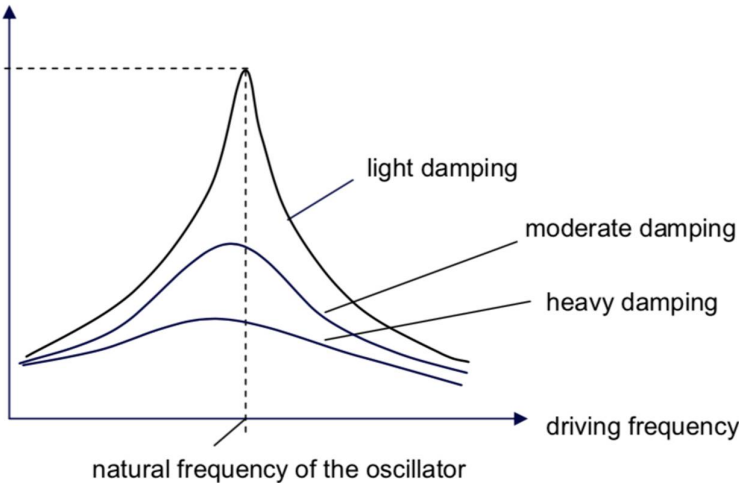


Question		Marks
2(a)	acceleration / force proportional to displacement (from a fixed point)	B1
	either acceleration and displacement in opposite directions or acceleration always directed towards a fixed point	B1
2(b)(i)1.	0.1 s, 0.3 s, 0.5 s, etc (any two)	B1
2(b)(i)2.	either 0 s, 0.4 s, 0.8 s, 1.2 s or 0.2 s, 0.6 s, 1.0 s (any two)	B1
2(b)(ii)	period = 0.4 s frequency = 2.5 Hz	B1
2(b)(iii)	phase difference = 90°	B1
2(c)	free: (body oscillates) without any loss of energy / no resistive forces / no external forces applied	B1
	forced: continuous energy input (required) / body is made to vibrate by an (external) periodic force / driving oscillator	B1
2(d)(i)	Amplitude of forced oscillations  Any of these shapes and max peak at natural frequency	B1
2(d)(ii)	frequency = 2.4 - 2.5 Hz	B1
2(d)(iii)	e.g. (1) attach sheet of card to trolley bigger than the cross section of the trolley increases damping / frictional force	M1 A1
	e.g. (2) reduce oscillator amplitude reduce power / energy input to system	M1 A1

Question	Marks
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3(a)	the work done per unit positive charge by an external agent in bringing a
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